

SAFe Strategic Plan

Driverless Software in Autonomous Vehicle Systems

1. Strategic Direction (Epic)

Epic Title: Achieve Market Leadership in Fully Autonomous Vehicle Software Systems

Timeline: 2025 – 2029 (2–4 Years)

Goal

Driverless Software will become the foremost provider of production ready and safety certified autonomous vehicle (AV) software by 2029. The company will deliver an end to end platform spanning perception, decision making and fleet management that enables OEM partners to deploy Level 4/5 autonomous vehicles at commercial scale.

Business Importance

Revenue Growth: Captures an estimated \$500B+ global AV software market projected by 2030.

Competitive Moat: First mover advantage in certified, production grade AV software creates high switching costs.

Pipeline: Leadership position attracts OEM, government and logistics partnerships.

Regulatory Readiness: Early compliance with ISO 26262 and FMVSS standards reduces time to market for clients.

2. Investment Themes

Per SAFe Portfolio Level Investment Themes translate strategy into budgetary priorities across program increments.

Theme	Description	Strategic Contribution
T1: Core AI & Perception	Invest in ML pipelines for real time sensor fusion eg.(LiDAR, radar and cameras).	Enables Level 4 autonomous decision making or product differentiator
T2: Safety & Compliance Engineering	Build ISO 26262 ASIL-D, SAE J3016, and GDPR-compliant safety frameworks and validation toolchains	Mandatory for commercial deployment; builds OEM trust and reduces liability
T3: Fleet Intelligence Platform	Develop cloud-based fleet monitoring, OTA update infrastructure and predictive analytics for AV operators.	Creates recurring SaaS revenue and supports scalable fleet deployments post launch

3. Feature Breakdown

Features are Program level requirements decomposed from the Epic. Each maps to a SAFe Agile Release Train (ART).

Feature F1: Autonomous Perception & Decision Engine (ART-1)

Develops the real time software stack for environment perception, path planning and motion control. Enables the vehicle to understand its surroundings and navigate without human input.

Contribution: Core technical foundation for all Level 4/5 autonomous functionality.

Investment Theme: T1: Core AI & Perception

ART Owner: Perception & Controls Engineering Team

Feature F2: Fleet Management & OTA Update Platform (ART-2)

Builds the cloud connected platform for real time fleet monitoring, remote diagnostics and secure over the air (OTA) software updates. Enables fleet operators to manage deployed AV units at scale.

Contribution: Unlocks post deployment revenue streams and ensures software currency across fleets.

Investment Theme: T3: Fleet Intelligence Platform

ART Owner: Platform Engineering & DevOps Team

4. User Stories

User stories follow the standard SAFe format: As a **[role]**, I want **[capability]**, so that **[benefit]**.

Feature F1: Autonomous Perception & Decision Engine

US-F1-01

Story: As an autonomous vehicle, I want to detect and classify pedestrians, cyclists and vehicles within a 360° sensor range so that I can avoid collisions in real time.

Task 1: Develop multiclass object detection model using YOLOv8 fine-tuned on AV datasets like (KITTI, nuScenes).

Task 2: Integrate LiDAR point cloud fusion with camera RGB data for depth classification.

US-F1-02

Story: As an autonomous vehicle, I want to generate a collision free path in real time so that I can navigate complex urban intersections without human intervention.

Task 1: Implement A* and RRT* path planning algorithms for dynamic obstacle avoidance.

Task 2: Validate planner response time against ASIL-D timing requirements which must be (< 50ms cycle).

US-F1-03

Story: As a safety engineer, I want all the decision making logic to be traceable to ISO 26262 ASIL-D requirements so that the software stack meets functional safety certification.

Task 1: Create requirements traceability matrix linking perception/planning modules to safety requirements.

Task 2: Run fault injection testing suite to validate fail safe behaviours under sensor failure.

US-F1-04

Story: As an OEM integration engineer, I want a documented API for the perception engine so that I can integrate Driverless Software into our existing vehicle ECU architecture.

Task 1: Define and publish RESTful and AUTOSAR compliant interfaces for perception outputs.

Task 2: Create integration test harness and sandbox environment for OEM testing.

US-F1-05

Story: As a test engineer, I want to run simulation-based regression tests on the perception stack so that I can validate software changes before deploying to physical vehicles.

Task 1: Set up CARLA/SUMO simulation environment with scripted test scenarios.

Task 2: Build CI/CD pipeline to trigger simulation regression on every pull request.

Feature F2: Fleet Management & OTA Update Platform

US-F2-01

Story: As a fleet manager, I want a real-time dashboard showing vehicle location, status, and health metrics so that I can proactively manage fleet operations.

Task 1: Design fleet dashboard UI with live telemetry feeds using WebSocket + React

Task 2: Build backend aggregation service to normalize telemetry from heterogeneous vehicle hardware

US-F2-02

Story: As a fleet operator, I want to push OTA software updates to vehicles remotely and schedule rollouts by zone so that I can minimize operational downtime.

Task 1: Implement delta-update mechanism with cryptographic signing (SHA-256 + RSA-2048).

Task 2: Build rollout orchestration service supporting staged, zone based and emergency update modes.

US-F2-03

Story: As a maintenance technician, I want to receive automated alerts when predictive diagnostics detect a potential hardware fault so that I can schedule proactive maintenance before failure.

Task 1: Train anomaly detection ML model on historical sensor telemetry for predictive fault detection.

Task 2: Configure alerting pipeline (email/SMS/PagerDuty) with severity classification.

US-F2-04

Story: As a compliance officer, I want all vehicle telemetry data stored with GDPR-compliant encryption and access controls so that the company meets regional data privacy regulations.

Task 1: Implement AES-256 encryption at rest and TLS 1.3 in transit for all telemetry data.

Task 2: Build RBAC access control layer with audit logging on all data access events.

US-F2-05

Story: As a product manager, I want usage analytics reports on vehicle performance KPIs so that I can make data-driven decisions on software prioritization.

Task 1: Define KPI schema (MTBF, disengagement rate, miles per incident) and implement ETL pipeline.

Task 2: Build self-serve analytics dashboard with configurable date/fleet filters.

5. Release Planning

SAFe releases align with Program Increments (PIs). Each PI spans 10 weeks (4 × 2 week sprints + 1 hardening sprint). Releases are delivered every 90 days per the Agile Release Train cadence.

Release 1: "Foundation" (PI-1 & PI-2 | Months 1–6)

Attribute	Details
Goal	Establish core perception engine baseline and cloud infrastructure; deliver internal demo-ready AV prototype.
Key Features	F1 (partial): Object detection, sensor fusion, path planner v1. F2 (partial): Telemetry ingestion, dashboard v1, OTA framework setup.
User Stories	US-F1-01, US-F1-02, US-F1-05 (simulation CI/CD), US-F2-01 (dashboard v1), US-F2-04 (data compliance).
Exit Criteria	Perception stack passes simulation regression suite with >95% object detection accuracy. Dashboard live with real vehicle telemetry.
Risks	Sensor hardware availability; LiDAR driver integration delays. Mitigation: use simulation fallback for blocked stories.

Release 2 — "Scale" (PI-3 & PI-4 | Months 7–12)

Attribute	Details
Goal	Deliver safety certified perception stack, production grade OTA platform and fleet management for pilot OEM partner deployment.
Key Features	F1 (complete): ASIL-D validation, OEM API, full fail-safe coverage. F2 (complete): OTA rollout, predictive diagnostics, analytics portal.
User Stories	US-F1-03 (safety traceability), US-F1-04 (OEM API), US-F2-02 (OTA), US-F2-03 (predictive alerts), US-F2-05 (analytics).
Exit Criteria	ISO 26262 ASIL-D pre-certification audit passed. OTA system deployed to pilot fleet (min. 25 vehicles). All user stories accepted by Product Owner.
Risks	Regulatory certification timelines. Mitigation: engage certification body (TÜV SÜD) from PI-3 start to allow parallel audit preparation.

6. SAFe Hierarchy Summary

SAFe Level	Artifact	Owner	Item
Portfolio	Epic	Portfolio Manager	Market Leadership in AV Software
Portfolio	Investment Themes	Portfolio Manager	T1: AI/Perception, T2: Safety, T3: Fleet Intelligence
Program	Features	Product Manager	F1: Perception Engine, F2: Fleet Platform
Program	Releases	Release Manager	Release 1: Foundation (M1–6), Release 2: Scale (M7–12)
Project	User Stories	Product Owner	10 user stories across F1 and F2
Project	Tasks	Development Team	2 tasks per user story (20 tasks total)